

Magnetic Double Helix

Yang Zhang^{✉*} and Paul M. Bellan[✉]

Applied Physics and Materials Science, California Institute of Technology, Pasadena, California 91125, USA



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Magnetic flux ropes, fundamental magnetohydrodynamic structures, often form braided helical configurations with net axial current, as observed in astrophysical and laboratory plasmas. However, an equilibrium model for such structures has remained elusive. We present a first-principles derivation of the braided flux ropes equilibrium by capturing the force balance of current-carrying plasma strands. This equilibrium reproduces the observed structure of the double helix nebula and provides a powerful tool for studying magnetic self-organization, plasma confinement, and astrophysical dynamics.

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Braided magnetic flux ropes have often been observed in solar and astrophysical contexts. Examples are the famous “granddaddy” prominence (see Fig. 1.1 of Ref. [1]), braided magnetic solar loop strands [2–4], double flux rope magnetic clouds embedded in the solar wind [5], the double helix nebula [6], and the helical structures of astrophysical jets [7,8].

Of special importance, Sun *et al.* [9] reported that in the Reconnection Scaling Experiment at Los Alamos National Labs, two braided magnetic flux ropes having same-direction axial current and wrapped around each other did not merge as would parallel-current straight wires; instead the two flux ropes moved closer together but then bounced back. This bouncing behavior indicated a more complex interaction than previously assumed and suggests that the wrapping of the flux ropes around each other results in an interaction different from that of two adjacent flux ropes which are not wrapped around each other.

Caltech solar loop experiments [10,11] produced braided loop structures having same-direction axial currents that also did not merge. During an instability cascade different strands of the Caltech braided structures behaved independently with sequential bursting of individual strands producing hard x-rays. Similar helical braided structures have also been observed in tokamaks [12], reversed field pinches [13], Z pinches [14], spheromaks [15], and plasma jets [16,17]. Marshall and Bellan [18] demonstrated that a braided structure is a lower-energy configuration than an unbraided structure having the same flux; this implies it is energetically favorable for flux-conserving configurations

to be braided. These observations of braided flux ropes appear to contradict a widely used model by Parker [19] where it was argued that two or more flux tubes wrapped around each other cannot be in equilibrium. This conflict between observations and a widely used model raises the critical question of whether it is possible to construct an equilibrium model for two flux ropes wrapped around each other, i.e., braided flux ropes.

The most basic braided flux rope system can be expressed in cylindrical coordinates (r, θ, z) as a configuration depending only on radial distance r and the helical parameter $u = \theta - kz$ where k is a wave number. Setting $u = \text{const}$ for fixed r generates a helical line in space, and all physical scalars in this helically symmetric system are invariant along this line. Johnson *et al.* [20] derived a helically symmetric magnetohydrodynamic (MHD) equilibrium differential equation known as the generalized Grad-Shafranov or JOKF equation. Because of its complexity, only solutions of the JOKF equation with no net axial current (alternating internal J_z polarity) have been reported [3,21] and these solutions are only obtained under restrictive linear assumptions. The derivation of an MHD helical equilibrium where all strands have the same J_z polarity, i.e., net axial current, has remained elusive. This is important because, unlike a no-net-axial-current configuration such a net-axial current configuration would be susceptible to kinking and associated expansion from hoop forces as observed in actual braided solar structures and in laboratory experiments.

We present here an MHD model that successfully produces an equilibrium of two braided magnetic flux ropes with $J_z \geq 0$ everywhere; the result of this model is shown in Fig. 1(a). This double helix equilibrium arises from two intertwined helical currents with a uniform current distribution inside each flux rope and zero current outside, as illustrated in Fig. 1(b). Within the current-carrying region, the plasma pressure gradient [see Fig. 1(c)] balances the $\mathbf{J} \times \mathbf{B}$ forces maintaining equilibrium.

*Contact author: yz0172@princeton.edu

Present address: Department of Astrophysical Sciences, Princeton University, Princeton, NJ 08544, USA and University Corporation for Atmospheric Research, Boulder, CO 80301, USA.

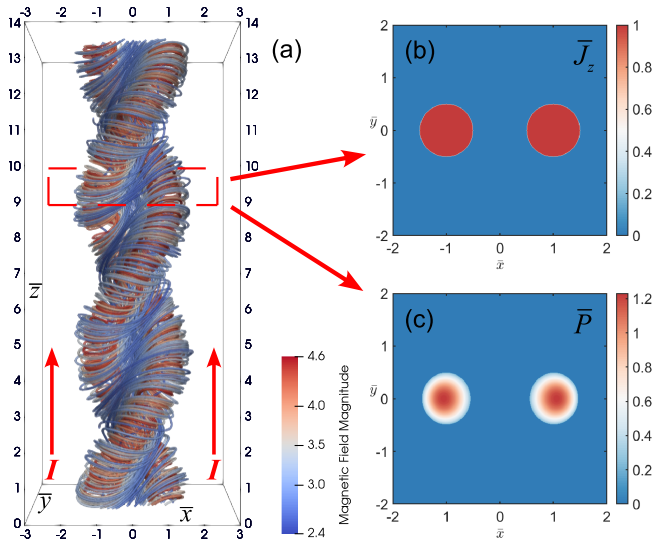


FIG. 1. Braided magnetic flux ropes under helical symmetry with the same direction current and axial wavelength $\bar{\lambda} = 9.17$. (a) 3D visualization of the magnetic field lines. The color represents the magnitude of the magnetic field $\bar{B} = \sqrt{\bar{B}_x^2 + \bar{B}_y^2 + \bar{B}_z^2}$. (b) The axial current density $\bar{J}_z = J_z/J_{z0}$ distribution on a given z slice at $\bar{z} = 9.17$. (c) The reconstructed pressure profile from $\nabla P = \mathbf{J} \times \mathbf{B}$.

This could, for example, represent a braided solar corona loop carrying a net axial current.

The mathematical details of the model are given in the Appendix, section on magnetic field under helical symmetry, and are summarized as follows: the magnetic field of a “point source” helical current wire is first established and then, via the Green’s function method, integration of a distribution of point-source magnetic fields is used to obtain the magnetic field of a distributed current. The braided flux ropes can be considered to be finite-diameter, plasma-filled, current-carrying “wires” that have finite internal pressure and are subject to both $\mathbf{J} \times \mathbf{B}$ forces and internal pressure gradient forces. The net magnetic force summed over the cross section of one of the two finite-cross-section helical wires is calculated from the current distribution in the wire and the magnetic field that was constructed using the Green’s function method. Equilibrium is found by determining the parameters for which this net force vanishes. This successfully reproduces the observed braided structure of the double helix nebula [6]. This model can be extended to represent other configurations of braided magnetic flux ropes by modifying the current distribution.

We now show via a thought experiment why it should be possible to construct a braided magnetic flux rope equilibrium where all strands have the same J_z polarity. Specifically, Fig. 2 shows two braided current wires having a range of axial angular wave numbers $k = 2\pi/\lambda$ where λ is the axial wavelength. When $\lambda \rightarrow \infty$ corresponding to

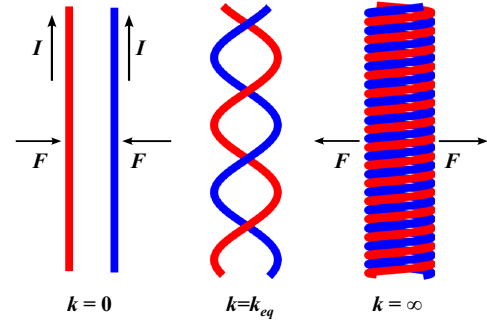


FIG. 2. A thought experiment on the interaction between two helically braided current wires under different angular wave numbers. The $k = 0$ corresponds to two parallel current wires. They attract each other. The $k = \infty$ corresponds to tightly coiled solenoids. The two helical wires repel each other. As the physics should be continuous, there must be a $k = k_{eq}$ as presented in the middle with a zero interaction force. This corresponds to the equilibrium state.

$k \rightarrow 0$, the two wires are parallel (left-hand side of Fig. 2). Since the two wires have the same-direction axial current, they attract each other. In contrast, when $k \rightarrow \infty$, the two wires form two interspersed tightly coiled same-direction solenoids, and the interaction force is now radially outward (right-hand side of Fig. 2). As the physics is continuous from $k = 0$ to $k = \infty$, there must be an intermediate state k_{eq} where the interaction force vanishes (middle of Fig. 2). This state represents the equilibrium state of the system.

The magnetic field generated from an arbitrary current distribution is derived in the Appendix and is given by Eqs. (A4)–(A6). This result can then be used to construct various braided flux rope situations by suitable choice of localized current density distribution $J_{z0}(r', \varphi')$, where r' and φ' are the radial and azimuthal coordinates in the source frame. Here we consider the simplest multistrand helical situation, namely two helically braided current wires each with a uniform current density over a circular cross section having a radius b in the $z = 0$ plane. The respective centers of these two circular cross sections are at $r = a$, $\theta = 0$ and at $r = a$, $\theta = \pi$, and through each circular cross section there is a uniform axial current density J_{z0} ; this is shown in Fig. 1(b) where $\bar{x} = x/a$, $\bar{y} = y/a$. The interiors of the two circular cross sections are defined by $(x \pm a)^2 + y^2 \leq b^2$. Since the wires are helical, the vector current density is $\mathbf{J} = J_{z0}[\hat{z} + kr'\hat{\theta}]$ inside the two circular cross sections while outside, $\mathbf{J} = 0$. The magnetic field resulting from this current distribution is derived in Supplemental Material [22] as Eqs. (S6)–(S8) using Eqs. (A4)–(A6). The area of each circular cross section in the $z = 0$ plane is $S = \pi b^2$.

The internal pressure distribution in a closed box cannot exert a net body force on the box because if this happened, the box would be imposing a force on its center of mass and so would be “pulling itself up by its own bootstraps.”

This restriction implies that $\int_S dS \nabla P = 0$, showing that the internal pressure distribution in a wire cannot exert a body force on the wire. Thus, the only possible net body force on each of the two wires in the double helix is the magnetic force, but in a static equilibrium there must be no net force so $\mathbf{f} = \int_S dS \mathbf{J} \times \mathbf{B} = 0$. Consequently, the equilibrium condition is that on each wire the net magnetic force per length is

$$\mathbf{f} = \int_S dS \mathbf{J} \times \mathbf{B} = 0. \quad (1)$$

Consider the magnetic force on the wire having a circular cross section in the $z = 0$ plane centered at $r = a$ and $\theta = 0$. The integration region S is $(x - a)^2 + y^2 \leq b^2$. Since each component of $\mathbf{B}$ is proportional to J_{z0} , each component of $\mathbf{f}$ is proportional to J_{z0}^2 and so is insensitive to the sign of J_{z0} . If it were assumed that $f_y > 0$ then, from symmetry, there would be a body force in the $+\hat{\theta}$ direction on both circular cross sections. This body force would cause the entire system to spin up in the $+\hat{\theta}$ direction and so gain angular momentum. Since the system is self-contained, it cannot change its own angular momentum and so f_y must vanish. A similar argument indicates that f_z must vanish. However, since f_x corresponds to the radial direction, a finite f_x would not change the center of mass velocity of the system. Thus, f_x could be finite, and, if so, would cause a radial expansion or contraction of the helix, i.e., a change in the radial separation between the two circular cross sections. Thus, there could only be a radial force between the two helical wires. A more detailed mathematical proof for $f_y = f_z = 0$ is presented in the Appendix section on magnetic force calculation. f_x is given by Eq. (A11).

To simplify the notation we define dimensionless lengths normalized to a , so $\bar{x} = x/a$, $\bar{b} = b/a$, $\bar{k} = ka$. We additionally define the dimensionless force $\bar{f}_x = f_x / (\mu_0 J_{z0}^2 \pi^2 b^4 / 4\pi a)$ and dimensionless magnetic field $\bar{\mathbf{B}} = \mathbf{B} / (\mu_0 J_{z0} b^2 / 4a)$. When $\bar{k} = 0$, $\bar{f}_x = -1$. This agrees with the attractive force between two parallel current wires each having a radius b and having centers separated by a distance $2a$ as this force is $\mu_0 J_{z0}^2 \pi^2 b^4 / 4\pi a$.

Figure 3(a) shows $\bar{f}_x$ plotted as a function of $\bar{k}$ for $\bar{b} = 0.5$. This plot shows that as $\bar{k}$ increases, $\bar{f}_x$ changes from being negative (attractive force) to being positive (repulsive force) as was shown qualitatively in Fig. 2. In particular, Fig. 3(a) shows that, for $\bar{b} = 0.5$, $\bar{f}_x = 0$ occurs at $\bar{k}_{\text{eq}} = 0.685$; this is the equilibrium. The value of $\bar{k}_{\text{eq}}$ at which equilibrium occurs depends on $\bar{b}$; this dependence is shown in Fig. 3(b). It is seen that $\bar{k}_{\text{eq}}$ is an increasing function of $\bar{b}$, indicating that the double helix equilibrium of thicker braided current wires (bigger $\bar{b}$) has a shorter axial wavelength.

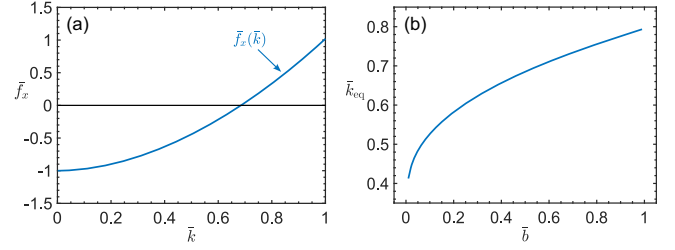


FIG. 3. Equilibrium of two helically braided current wires. (a) $\bar{f}_x$ versus $\bar{k}$ under $\bar{b} = 0.5$. $\bar{f}_x = 0$ corresponds to the equilibrium state, and $\bar{k}_{\text{eq}} = 0.685$. (b) The equilibrium axial wave number $\bar{k}_{\text{eq}}$ versus minor radius $\bar{b}$.

Figure 4 provides a detailed visualization of the physical characteristics of the equilibrium state for a helically braided magnetic flux rope system with $\bar{b} = 0.5$ and $\bar{k}_{\text{eq}} = 0.685$. It shows various physical quantities in the $\bar{z} = 0$ plane; physical quantities at different $\bar{z}$ can be determined from $\bar{z} = 0$ solutions utilizing the helical symmetry. Figure 4(a) shows the axial current density $\bar{J}_z = J_z / J_{z0}$ while Fig. 4(b) shows the azimuthal current density $\bar{J}_\theta = J_\theta / J_{z0}$. Figure 4(c) shows $\bar{\mathbf{B}}_\perp$, the magnetic field component perpendicular to $\hat{z}$, as lines following $\bar{\mathbf{B}}_\perp = \bar{B}_x \hat{x} + \bar{B}_y \hat{y}$; it is seen that the $\bar{\mathbf{B}}_\perp$ lines encircle the two current centers. If the perpendicular magnetic field component is written as $\bar{\mathbf{B}}_\perp = \bar{B}_r \hat{r} + \bar{B}_\theta \hat{\theta}$, it is seen that $\bar{B}_r$ is an odd function of θ and $\bar{B}_\theta$ is an even function of θ . The colors in Fig. 4(c) show the amplitude $\bar{B}_\perp = \sqrt{\bar{B}_x^2 + \bar{B}_y^2}$. Figure 4(d) shows $\bar{B}_z$ which is largest at radii internal to the

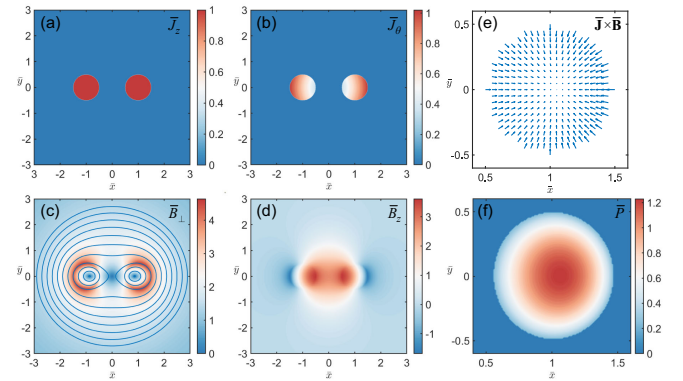


FIG. 4. Braided double helix equilibrium with $\bar{b} = 0.5$ and $\bar{k}_{\text{eq}} = 0.685$. (a) The axial current density $\bar{J}_z = J_z / J_{z0}$ distribution. The axial current density is uniform inside the circular, current-carrying cross-section. (b) The azimuthal current density $\bar{J}_\theta = J_\theta / J_{z0}$ distribution. (c) The $x - y$ direction magnetic field lines $\bar{\mathbf{B}}_\perp = \bar{B}_x \hat{x} + \bar{B}_y \hat{y}$ and, as colored shading, corresponding amplitude $\bar{B}_\perp = \sqrt{\bar{B}_x^2 + \bar{B}_y^2}$. (d) The axial magnetic field $\bar{B}_z$ distribution. (e) Magnetic force $\mathbf{J} \times \mathbf{B}$ inside of one current wire on $\bar{z} = 0$ plane. (f) The reconstructed pressure profile from $\nabla P = \mathbf{J} \times \mathbf{B}$.

braided helical wires and reverses direction on moving from inside to outside the helix. It is seen that $\bar{B}_z$ is an even function of θ . These symmetries of $\bar{B}_r$, $\bar{B}_\theta$, and $\bar{B}_z$ justify the conclusion that $f_y = f_z = 0$. The equilibrium is determined from the surface integral of the magnetic field force equaling zero, i.e., $f_x = 0$. Inside the circular current-carrying cross section, the local magnetic force $\mathbf{J} \times \mathbf{B}$ can be finite, as shown in Fig. 4(e), and this force acts to pinch the current wire. This nonzero magnetic force can be balanced by an internal pressure gradient ∇P . Integration of $\nabla P = \mathbf{J} \times \mathbf{B}$ gives an internal pressure profile in the wire $\bar{P} = P/(\mu_0 J_{z0}^2 b^2/4)$. This internal pressure profile is shown in Fig. 4(f). The pressure is obtained by integrating the magnetic force $\mathbf{J} \times \mathbf{B}$ inward from the boundary, where the pressure is assumed to be zero. The equilibrium $\nabla P = \mathbf{J} \times \mathbf{B}$ requires $\nabla \times (\mathbf{J} \times \mathbf{B}) = 0$. The magnetic field generated from the double helix current wires automatically satisfies this curl-free condition $\nabla \times (\mathbf{J} \times \mathbf{B}) = 0$; details are discussed in Supplemental Material [22]. The 3D visualization of the magnetic field lines is presented in Fig. 1(a). The field lines wrap around the two current-carrying wires forming a double helix magnetic structure, so the magnetic field generated by braided current wires is also braided.

We have thus provided a systematic method for determining the equilibrium of helically braided systems. This method is straightforward to implement, involving several key steps: (i) define a current distribution function to represent the helically braided system, (ii) calculate the magnetic field using Eqs. (A4)–(A6), (iii) determine the equilibrium state by adjusting the normalized wave number ka to obtain zero magnetic force integrated over the cross section of one current carrying wire, and (iv) calculate the internal pressure in the plasma-filled flux rope from $\nabla P = \mathbf{J} \times \mathbf{B}$. An important corollary of this method is that it could be used to construct a fractally braided system analogous to a braided rope where each rope braid is composed of strands that, in turn, are composed of smaller braided strands.

The braided helical current wire configuration is more stable than the unbraided, straight wire configuration because the braiding generates an axial magnetic field at the center. When a perturbation occurs in the global braided structure, this axial magnetic field provides a restoring force that helps stabilize the system. In Supplemental Material [22], we present a detailed stability analysis [23] of a simplified braided helical current system. The analysis demonstrates that such a system is stable for the $m = 0$ sausage mode and $m \geq 2$ modes, and also stable for the $m = 1$ kink mode, as long as the axial length of the perturbation is smaller than $2.2\lambda_{eq}$, where λ_{eq} is the braid pitch length. This analysis does not involve any external magnetic field or metal wall, similar to conditions found in astrophysical structures. These findings may provide a plausible explanation for the existence of elongated braided

structures, such as the double helix nebula [6] and astrophysical jets [7], in astrophysics.

We can reproduce observed helically braided structures using the double helix equilibrium derived from this helical current two-wire model. A specific example is the braided structure of the double helix nebula [6], shown in Fig. 5(a). It has highly ordered magnetic field with synchrotron-emitting relativistic electrons [24]. The minor radius is measured as $\bar{b} \approx 0.3$, and the axial angular wave number $\bar{k} \approx 0.6$. We assume that the two wires have the same direction current and that this current is uniformly distributed on a circular cross section. Using $\bar{b} = 0.3$, the helical wire model yields an equilibrium $\bar{k} = 0.623$, which is in excellent agreement with the observed value. Figure 5(b) shows the double helix structure predicted by the helical wire model presented here oriented as in the telescope photo and shows that this closely replicates the intricate structure of the double helix nebula in Fig. 5(a). Morris *et al.* [6] proposed that this double helix structure is caused by torsional Alfvén waves. However, torsional Alfvén waves do not generate a collective motion of the whole flux rope that modulates the center axis [25,26]. From the matching of the equilibrium, the observed helical structure is more plausibly interpreted as two current-carrying magnetic flux ropes braided around each other, forming an equilibrium, as in Fig. 5(b). The overall length of the double helix nebula is $1.5\lambda_{eq}$ and so is less than $2.2\lambda_{eq}$, indicating that this double helix equilibrium is MHD stable.

An important question regarding the equilibrium of braided magnetic flux ropes is the return current path. Possible configurations for this current are sketched in Fig. S1 of Supplemental Material [22].

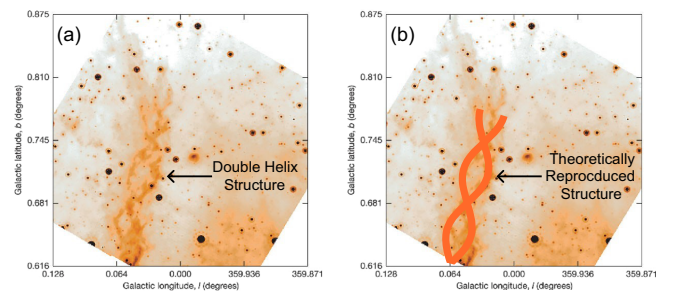


FIG. 5. Comparison between the double helix nebula and a double helix equilibrium reconstructed from the model. (a) The double helix nebula, observed at the infrared wavelength of $24\ \mu\text{m}$ with the multiband imaging photometer for Spitzer (MIPS) camera on the Spitzer Space Telescope. The spatial resolution is 6 arcsec. At the 8 kpc distance of the Galactic Center, 1 arc min corresponds to 2.5 pc. The minor radius is measured as $\bar{b} \approx 0.3$, and the axial angular wave number $\bar{k} \approx 0.6$. This image is reproduced from [6]. (b) A double helix equilibrium reconstructed from the model with $\bar{b} = 0.3$ and $\bar{k}_{eq} = 0.623$.

In some braided magnetic flux rope experiments, an external axial magnetic field was applied, and the rotation of two flux ropes was observed [9,27]. The helical current wires model can incorporate axial magnetic field and rotation. The rotation provides an additional radial force to the current wire, which can be considered as an extra force in the force balance equation. In astrophysical jet observations, rotating flows were also observed [28]. This model can also be expanded to a dynamic equilibrium with helical flow through the structure. Examples are provided in Supplemental Material [22].

In summary, this Letter presents a comprehensive helical current wire model that establishes MHD equilibria. The equilibria correlate with phenomena observed in both astrophysical environments and controlled laboratory experiments. The application of this equilibrium model extends broadly to plasma studies, including but not limited to the reconstruction of magnetic flux ropes [29], magnetic flux rope interactions [30] and evolutions [31,32], the study of magnetic helicity [1], magnetic reconnections [33,34], and magnetic fusion equilibrium states [35].

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Data availability—The data and code that support the findings of this Letter are openly available [37].

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End Matter

Appendix: Magnetic field derivation—Magnetic field under helical symmetry: To derive the Green’s function for the magnetic field produced by distributed helical currents, we begin by considering an infinitely long wire that follows a helical path having radius r' and axial wave number k . The wire has zero radius and carries a current I and so has a current density $\mathbf{J}(r, \theta, z) = Ir^{-1}\delta(r - r')\delta(\theta - kz - \varphi')(\hat{z} + kr'\hat{\theta})$. At $z = 0$ the wire is located at azimuthal position φ' ; this dependence on r' and φ' enables a later construction having multiple wires where each has a different r' and φ' . The wire thus is at $r(z) = r'$, $\theta(z) = kz + \varphi'$.

Since the regions $r < r'$ (i.e., interior to the wire) and $r > r'$ (i.e., exterior to the wire) are both vacuum, the magnetic field in both the interior and exterior regions

satisfies $\nabla \times \mathbf{B} = 0$, so in both these regions $\mathbf{B} = \nabla\psi$ where ψ satisfies Laplace’s equation since $\nabla \cdot \mathbf{B} = 0$. However, the interior and exterior ψ differ with a jump in the character of ψ at $r = r'$ determined by the electric current in the wire. Using these properties, the magnetic field is derived via the discontinuous vacuum magnetic field method in Tominaka *et al.* [36]. These give $\mathbf{B}_{\text{interior}} = \nabla\psi_{\text{interior}}$ and $\mathbf{B}_{\text{exterior}} = \nabla\psi_{\text{exterior}}$. At $r = r'$ the radial component of the magnetic field is continuous, but there are jumps in B_θ and B_z corresponding to the current in the wire. The azimuthal delta function character of the wire results in there being an infinite sum of harmonic functions.

On using these relations, Tominaka *et al.* [36] showed that for $r < r'$ the magnetic field produced by the wire is

$$\begin{aligned}
 B_r(r, \theta, z) &= \frac{\mu_0 I}{\pi} k^2 r' \sum_{n=1}^{\infty} n K'_n(nkr') I'_n(nkr) \sin[n(\theta - kz - \varphi')], \\
 B_\theta(r, \theta, z) &= \frac{\mu_0 I}{\pi} kr' \frac{1}{r} \sum_{n=1}^{\infty} n K'_n(nkr') I_n(nkr) \cos[n(\theta - kz - \varphi')], \\
 B_z(r, \theta, z) &= \frac{\mu_0 I}{2\pi} k - \frac{\mu_0 I}{\pi} k^2 r' \sum_{n=1}^{\infty} n K'_n(nkr') I_n(nkr) \cos[n(\theta - kz - \varphi')],
 \end{aligned} \tag{A1}$$

while for $r > r'$ the magnetic field is given by

$$\begin{aligned} B_r(r, \theta, z) &= \frac{\mu_0 I}{\pi} k^2 r' \sum_{n=1}^{\infty} n I'_n(nkr') K'_n(nkr) \sin[n(\theta - kz - \varphi')], \\ B_\theta(r, \theta, z) &= \frac{\mu_0 I}{2\pi r} + \frac{\mu_0 I}{\pi} k r' \frac{1}{r} \sum_{n=1}^{\infty} n I'_n(nkr') K_n(nkr) \cos[n(\theta - kz - \varphi')], \\ B_z(r, \theta, z) &= -\frac{\mu_0 I}{\pi} k^2 r' \sum_{n=1}^{\infty} n I'_n(nkr') K_n(nkr) \cos[n(\theta - kz - \varphi')]. \end{aligned} \quad (\text{A2})$$

Here $I_n(nkr)$ is the modified Bessel function of the first kind and $K_n(nkr)$ is the modified Bessel function of the second kind. $I'_n(nkr)$ and $K'_n(nkr)$ are derivatives with respect to the function arguments. This magnetic field has the properties that (i) B_r is continuous at $r = r'$, (ii) there is an axial field $B_z = (\mu_0 I / 2\pi) k$ for $r \ll r'$ as befits a radius r' solenoid wound of a wire with pitch $2\pi/k$, (iii) for $r \gg r'$ there is an azimuthal field $B_\theta = (\mu_0 I / 2\pi r)$ as befits an axial wire with current I , (iv) there are jumps in B_θ and B_z at $r = r'$ corresponding to the current in the wire, and (v) the field is potential for $r < r'$ and for $r > r'$. The magnetic field prescribed by Eqs. (A1) and (A2) is from a point-

source helical current. Using the Green's function method, we obtain the magnetic field of a distributed current by replacing the current I with an area integration over the current density of the distributed current. Because the current flowing in the helical wire can be completely specified by prescribing J_z in the $z = 0$ plane, we define $J_{z0}(r', \varphi') = J_z(r', \varphi' - kz, z = 0)$. We now consider a thin helical wire with current $dI = J_{z0}(r', \varphi') r' dr' d\varphi'$; this current flows through an annular sector in the $z' = 0$ plane located between r' and $r' + dr'$ and between φ' and $\varphi' + d\varphi'$. Using Eqs. (A1) and (A2), the radial magnetic field provided by this current dI is

$$dB_r(r, \theta, z) = \begin{cases} \frac{\mu_0 J_{z0}(r', \varphi') r' dr' d\varphi'}{\pi} k^2 r' \sum_{n=1}^{\infty} n K'_n(nkr') I'_n(nkr) \sin[n(\theta - kz - \varphi')] & \text{for } r < r' \\ \frac{\mu_0 J_{z0}(r', \varphi') r' dr' d\varphi'}{\pi} k^2 r' \sum_{n=1}^{\infty} n I'_n(nkr') K'_n(nkr) \sin[n(\theta - kz - \varphi')] & \text{for } r > r'. \end{cases} \quad (\text{A3})$$

The radial magnetic field generated from a helical current distribution is then the integral of Eq. (A3) over all (r', φ') to give

$$\begin{aligned} B_r(r, \theta, z) &= \int_r^\infty dr' \int_0^{2\pi} d\varphi' \left\{ \frac{\mu_0 J_{z0}(r', \varphi')}{\pi} k^2 r'^2 \sum_{n=1}^{\infty} n K'_n(nkr') I'_n(nkr) \sin[n(\theta - kz - \varphi')] \right\} \\ &+ \int_0^r dr' \int_0^{2\pi} d\varphi' \left\{ \frac{\mu_0 J_{z0}(r', \varphi')}{\pi} k^2 r'^2 \sum_{n=1}^{\infty} n I'_n(nkr') K'_n(nkr) \sin[n(\theta - kz - \varphi')] \right\}. \end{aligned} \quad (\text{A4})$$

Similarly, the azimuthal magnetic field and the axial magnetic field can be derived as

$$\begin{aligned} B_\theta(r, \theta, z) &= \int_r^\infty dr' \int_0^{2\pi} d\varphi' \left\{ \frac{\mu_0 J_{z0}(r', \varphi')}{\pi} k r'^2 \frac{1}{r} \sum_{n=1}^{\infty} n K'_n(nkr') I_n(nkr) \cos[n(\theta - kz - \varphi')] \right\} \\ &+ \int_0^r dr' \int_0^{2\pi} d\varphi' \left\{ \frac{\mu_0 J_{z0}(r', \varphi') r'}{2\pi r} + \frac{\mu_0 J_{z0}(r', \varphi')}{\pi} k r'^2 \frac{1}{r} \sum_{n=1}^{\infty} n I'_n(nkr') K_n(nkr) \cos[n(\theta - kz - \varphi')] \right\}, \end{aligned} \quad (\text{A5})$$

$$\begin{aligned} B_z(r, \theta, z) &= -\int_0^r dr' \int_0^{2\pi} d\varphi' \left\{ \frac{\mu_0 J_{z0}(r', \varphi')}{\pi} k^2 r'^2 \sum_{n=1}^{\infty} n I'_n(nkr') K_n(nkr) \cos[n(\theta - kz - \varphi')] \right\} \\ &+ \int_r^\infty dr' \int_0^{2\pi} d\varphi' \left\{ \frac{\mu_0 J_{z0}(r', \varphi') r'}{2\pi} k - \frac{\mu_0 J_{z0}(r', \varphi')}{\pi} k^2 r'^2 \sum_{n=1}^{\infty} n K'_n(nkr') I_n(nkr) \cos[n(\theta - kz - \varphi')] \right\}. \end{aligned} \quad (\text{A6})$$

We perform the integration explicitly rather than invoking the Green's function approach directly, to make the derivation easy to follow. To strictly follow the Green's function method, one would first identify the response to a point current [given by Eqs. (A1) and (A2) normalized by I], then integrate over the source distribution. This yields the same result as our direct approach. Different localized current density distribution $J_{z0}(r', \varphi')$ can be constructed to model various braided flux rope situations. The magnetic field generated by a double helix current wire system is calculated in Supplemental Material [22].

Magnetic force calculation: For the magnetic force in Eq. (1), the integration region S is $(x-a)^2 + y^2 \leq b^2$, so $\int dS = \int_{a-b}^{a+b} r dr \int_{-c(r)}^{c(r)} d\theta$, where $c(r)$ is defined in Supplemental Material, as Eq. (S1). The three components of $\mathbf{f}$ are

$$f_x = \int_S dS J_{z0} [(krB_z - B_\theta) \cos \theta - B_r \sin \theta] \quad (\text{A7})$$

$$f_y = \int_S dS J_{z0} [(krB_z - B_\theta) \sin \theta + B_r \cos \theta], \quad (\text{A8})$$

$$f_z = \int_S dS J_{z0} [-krB_r]. \quad (\text{A9})$$

From Eqs. (S5)–(S7), B_r and $(krB_z - B_\theta) \sin \theta + B_r \cos \theta$ are an odd function of θ , so $f_y = 0$ and $f_z = 0$. Using Eq. (A7), f_x can be calculated as

$$f_x = \int_{a-b}^{a+b} r dr \int_{-c(r)}^{c(r)} d\theta J_{z0} [(krB_z - B_\theta) \cos \theta - B_r \sin \theta] \quad (\text{A10})$$

We define the dimensionless force $\bar{f}_x = f_x / (\mu_0 J_{z0}^2 \pi^2 b^4 / 4\pi a)$ and then use Eqs. (S5)–(S7) in Eq. (A10) to obtain

$$\bar{f}_x = \frac{32a}{\pi^2 k^3 b^4} \left\{ \begin{aligned} & \frac{k^3}{2} \int_{a-b}^{a+b} dr \left(- \int_{a-b}^r dr' r' c(r') + k^2 r^2 \int_r^{a+b} dr' r' c(r') \right) \sin c(r) \\ & - \sum_{i=1}^{\infty} \int_{nk(a-b)}^{nk(a+b)} dR \left\{ \begin{aligned} & \frac{n \cos c(\frac{R}{nk}) \sin nc(\frac{R}{nk}) - \sin c(\frac{R}{nk}) \cos nc(\frac{R}{nk})}{n^4(n^2-1)} \left(1 + \left(\frac{R}{n} \right)^2 \right) W_n(R) \\ & + \frac{\cos c(\frac{R}{nk}) \sin nc(\frac{R}{nk}) - n \sin c(\frac{R}{nk}) \cos nc(\frac{R}{nk})}{n^4(n^2-1)} \frac{R}{n} Q_n(R) \end{aligned} \right\} \end{aligned} \right\} \quad (\text{A11})$$

where $W_n(R)$ and $Q_n(R)$ are defined in Supplemental Material [22], as Eqs. (S2) and (S3). The dimensionless force $\bar{f}_x$ is calculated numerically, and the summation on n stops when the relative error $|\Delta \bar{f}_x / \bar{f}_x| < 10^{-5}$.